

Experiment No - 6

Aim: Apply Classification data mining technique on sample data sets in WEKA.

Date:

Competency and Practical Skills: Exploration and Understanding of Tool.

Relevant CO: CO4 & CO5

Objectives: 1) improve users' understanding of classification techniques 2) Familiarize with the tool

Equipment/Instruments: WEKA Tool

Safety and necessary Precautions:

Properly evaluate the classification rules and avoid drawing incorrect conclusions.

Procedure:

1. Install and set up WEKA.
2. Import your dataset.
3. Apply Classification data mining technique .

OUTPUT:

=== Run information ===

Scheme: weka.classifiers.trees.RandomForest -P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 S 1

Relation: heart-weka.filters.unsupervised.attribute.NumericToNominal-Rfirst-last

Instances: 1025 Attributes: 14 age sex cp trestbps
chol fbs restecg thalach exang oldpeak
slope ca thal
target

Test mode: split 66.0% train, remainder test

=== Classifier model (full training set) ===

RandomForest

Bagging with 100 iterations and base learner

weka.classifiers.trees.RandomTree -K 0 -M 1.0 -V 0.001 -S 1 -do-not-check-capabilities

Time taken to build model: 0.22 seconds

==== Evaluation on test split ====

Time taken to test model on test split: 0.04 seconds

==== Summary ====

Correctly Classified Instances	345	99.1379 %
Incorrectly Classified Instances	3	0.8621 %
Kappa statistic	0.9827	
Mean absolute error	0.1015	
Root mean squared error	0.1377	
Relative absolute error	20.32 %	
Root relative squared error	27.5542 %	
Total Number of Instances	348	

==== Detailed Accuracy By Class ====

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.016	0.982	1.000	0.991	0.983	1.000	1.000	0
	0.984	0.000	1.000	0.984	0.992	0.983	1.000	1.000	1
Weighted Avg.	0.991	0.008	0.992	0.991	0.991	0.983	1.000	1.000	

==== Confusion Matrix ====

```

a  b  <-- classified as
164  0 |  a = 0
3 181 |  b = 1

```

Demonstration of Tool:

WEKA tool was used to apply classification algorithm (J48).

The dataset was loaded and processed.

The model was trained using 10-fold cross-validation. The output included accuracy and confusion matrix.

Conclusion:

Classification technique was successfully implemented using WEKA. The model achieved good accuracy and correctly classified most instances. WEKA provides an easy interface for applying machine learning techniques.

Quiz:

1) What is classification and how can it be performed using WEKA tool?

Ans: Classification is a process of predicting class labels using trained data. In WEKA, it is performed using algorithms like J48, Naïve Bayes.

2) What are the key evaluation metrics used to assess the accuracy of a model in WEKA? Ans:

- Accuracy
- Precision
- Recall
- Confusion Matrix

Suggested Reference:

1. J. Han, M. Kamber, "Data Mining Concepts and Techniques", Morgan Kaufmann
2. <https://waikato.github.io/weka-wiki/documentation/>

References used by the students:

1. <https://waikato.github.io/weka-wiki/documentation/>

Rubric wise marks obtained:

Rubrics	Knowledge (2)		Problem Recognition (2)		Tool Usage (2)		Demonstration (2)		Ethics (2)		Total
	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	
Marks											

Experiment No - 7

Aim: 7.1.Implement Classification technique with quality Measures in any Programming language.
7.2 Implement Regression technique in any Programming language.

Date:

Competency and Practical Skills: Logic building, Programming and Analyzing

Relevant CO: CO5

Objectives:

(a) To evaluate the quality of the classification model using accuracy and confusion matrix. (b) To evaluate regression model

Equipment/Instruments: open-source software for programming

Theory:

Classification

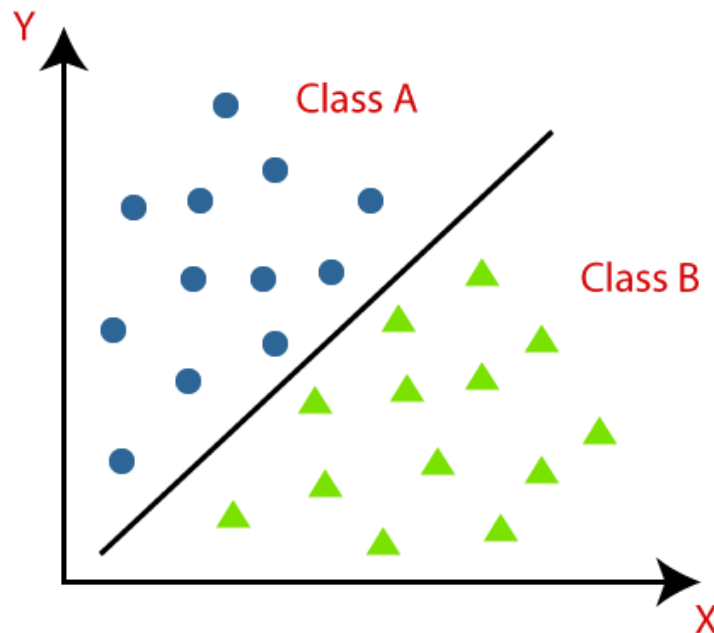
The Classification algorithm is a Supervised Learning technique that is used to identify the category of new observations on the basis of training data. In Classification, a program learns from the given dataset or observations and then classifies new observation into a number of classes or groups. Such as, **Yes or No, 0 or 1, Spam or Not Spam, cat or dog**, etc. Classes can be called as targets/labels or categories.

Unlike regression, the output variable of Classification is a category, not a value, such as "Green or Blue", "fruit or animal", etc. Since the Classification algorithm is a Supervised learning technique, hence it takes labeled input data, which means it contains input with the corresponding output. In classification algorithm, a discrete output function(y) is mapped to input variable(x).

$y=f(x)$, where y = categorical output

The best example of an ML classification algorithm is **Email Spam Detector**.

The main goal of the Classification algorithm is to identify the category of a given dataset, and these algorithms are mainly used to predict the output for the categorical data. Classification algorithms can be better understood using the below diagram. In the below diagram, there are two classes, class A and Class B. These classes have features that are similar to each other and dissimilar to other classes.



The algorithm which implements the classification on a dataset is known as a classifier. There are two types of Classifications:

Binary Classifier:

If the classification problem has only two possible outcomes, then it is called as Binary Classifier.

Examples:

YES or NO, MALE or FEMALE, SPAM or NOT SPAM, CAT or DOG, etc.

Multi-class Classifier:

If a classification problem has more than two outcomes, then it is called as Multi-class Classifier.

Example: Classifications of types of crops, Classification of types of music.

Learners in Classification Problems: In the classification problems, there are two types of learners:

Lazy Learners: Lazy Learner firstly stores the training dataset and wait until it receives the test dataset. In Lazy learner case, classification is done on the basis of the most related data stored in the training dataset. It takes less time in training but more time for predictions.

Example: K-NN algorithm, Case-based reasoning

Eager Learners: Eager Learners develop a classification model based on a training dataset before receiving a test dataset. Opposite to Lazy learners, Eager Learner takes more time in learning, and less time in prediction.

Example: Decision Trees, Naïve Bayes, ANN.

Types of ML Classification Algorithms:

Classification Algorithms can be further divided into the Mainly two category:

Linear Models

Logistic Regression

Support Vector Machines **Non-linear**

Models

K-Nearest Neighbours

Kernel SVM

Naïve Bayes

Decision Tree Classification

Random Forest Classification

Evaluating a Classification model:

Once our model is completed, it is necessary to evaluate its performance; either it is a Classification or Regression model. So for evaluating a Classification model, we have the following ways:

1. Log Loss or Cross-Entropy Loss:

It is used for evaluating the performance of a classifier, whose output is a probability value between the 0 and 1.

For a good binary Classification model, the value of log loss should be near to 0.

The value of log loss increases if the predicted value deviates from the actual value.

The lower log loss represents the higher accuracy of the model.

2. Confusion Matrix:

The confusion matrix provides us a matrix/table as output and describes the performance of the model.

It is also known as the error matrix.

The matrix consists of predictions result in a summarized form, which has a total number of correct predictions and incorrect predictions. The matrix looks like as below table:

	Actual Positive	Actual Negative
Predicted Positive	True Positive	False Positive
Predicted Negative	False Negative	True Negative

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total Population}}$$

3. AUC-ROC curve:

ROC curve stands for **Receiver Operating Characteristics Curve** and AUC stands for **Area Under the Curve**.

It is a graph that shows the performance of the classification model at different thresholds. To visualize the performance of the multi-class classification model, we use the AUC-ROC Curve. The ROC curve is plotted with TPR and FPR, where TPR (True Positive Rate) on Y-axis and FPR (False Positive Rate) on X-axis.

Use cases of Classification Algorithms

Classification algorithms can be used in different places. Below are some popular use cases of Classification Algorithms:

Email Spam Detection

Speech Recognition

Identifications of Cancer tumor cells.

Drugs Classification

Biometric Identification, etc.

Regression

Regression analysis is a statistical method to model the relationship between a dependent (target) and independent (predictor) variables with one or more independent variables. More specifically, Regression analysis helps us to understand how the value of the dependent variable is changing corresponding to an independent variable when other independent variables are held fixed. It predicts continuous/real values such as temperature, age, salary, price, etc.

We can understand the concept of regression analysis using the below example:

Example: Suppose there is a marketing company A, who does various advertisement every year and get sales on that. The below list shows the advertisement made by the company in the last 5 years and the corresponding sales:

Advertisement	Sales
\$90	\$1000
\$120	\$1300
\$150	\$1800
\$100	\$1200
\$130	\$1380
\$200	??

Now, the company wants to do the advertisement of \$200 in the year 2019 and wants to know the prediction about the sales for this year. So to solve such type of prediction problems in machine learning, we need regression analysis. Regression is a supervised learning technique which helps in finding the correlation between variables and enables us to predict the continuous output variable based on the one or more predictor variables. It is mainly used for prediction, forecasting, time series modeling, and determining the causaleffect relationship between variables.

In Regression, we plot a graph between the variables which best fits the given datapoints, using this plot, the machine learning model can make predictions about the data. In simple words, *"Regression shows a line or curve that passes through all the datapoints on targetpredictor graph in such a way that the vertical distance between the datapoints and the regression line is minimum."* The distance between datapoints and line tells whether a model has captured a strong relationship or not.

Some examples of regression can be as:

- Prediction of rain using temperature and other factors
- Determining Market trends
-

Prediction of road accidents due to rash driving. Terminologies Related to the Regression Analysis:

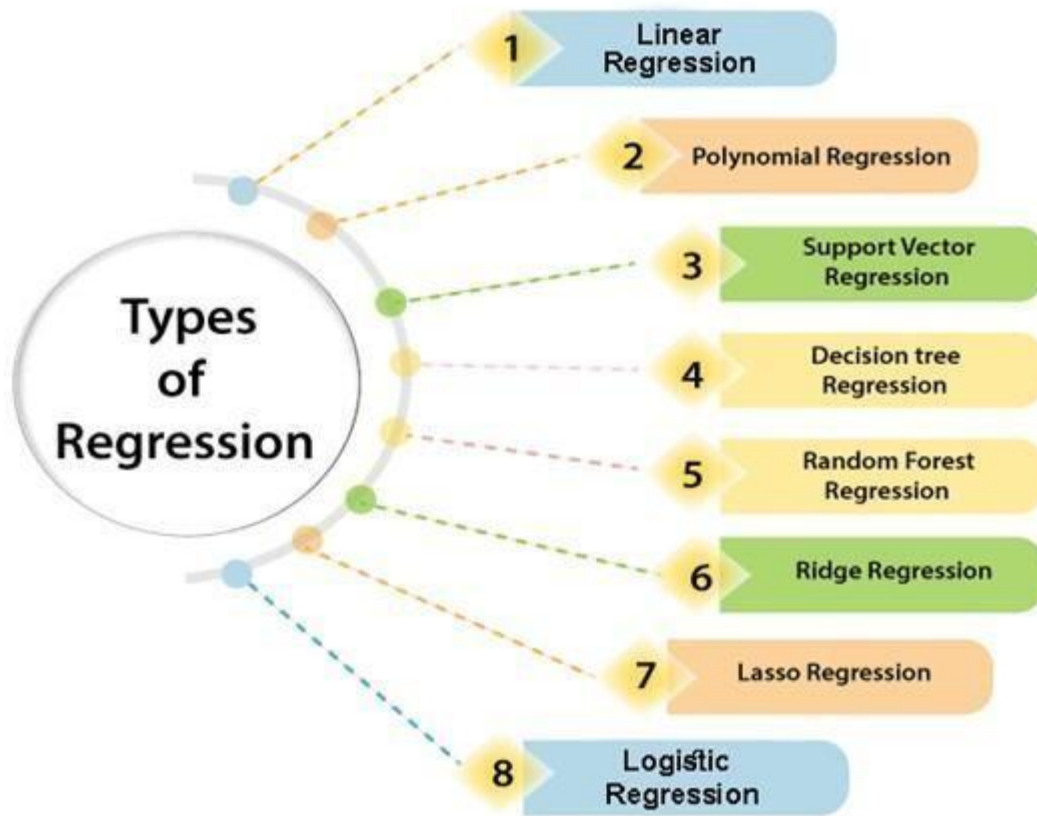
- Dependent Variable: The main factor in Regression analysis which we want to predict or understand is called the dependent variable. It is also called target variable.

- Independent Variable: The factors which affect the dependent variables or which are used to predict the values of the dependent variables are called independent variable, also called as a predictor.
- Outliers: Outlier is an observation which contains either very low value or very high value in comparison to other observed values. An outlier may hamper the result, so it should be avoided.
- Multicollinearity: If the independent variables are highly correlated with each other than other variables, then such condition is called Multicollinearity. It should not be present in the dataset, because it creates problem while ranking the most affecting variable.
- Underfitting and Overfitting: If our algorithm works well with the training dataset but not well with test dataset, then such problem is called Overfitting. And if our algorithm does not perform well even with training dataset, then such problem is called underfitting.
- Regression analysis helps in the prediction of a continuous variable. There are various scenarios in the real world where we need some future predictions such as weather condition, sales prediction, marketing trends, etc., for such case we need some technology which can make predictions more accurately.
- Regression estimates the relationship between the target and the independent variable.
- It is used to find the trends in data.
- It helps to predict real/continuous values.
- By performing the regression, we can confidently determine the most important factor, the least important factor, and how each factor is affecting the other factors.

Types of Regression

There are various types of regressions which are used in data science and machine learning. Each type has its own importance on different scenarios, but at the core, all the regression methods analyze the effect of the independent variable on dependent variables. Here we are discussing some important types of regression which are given below:

- Linear Regression
- Logistic Regression
- Polynomial Regression
- Support Vector Regression
- Decision Tree Regression
- Random Forest Regression
- Ridge Regression
- Lasso Regression

**Safety and necessary Precautions:**

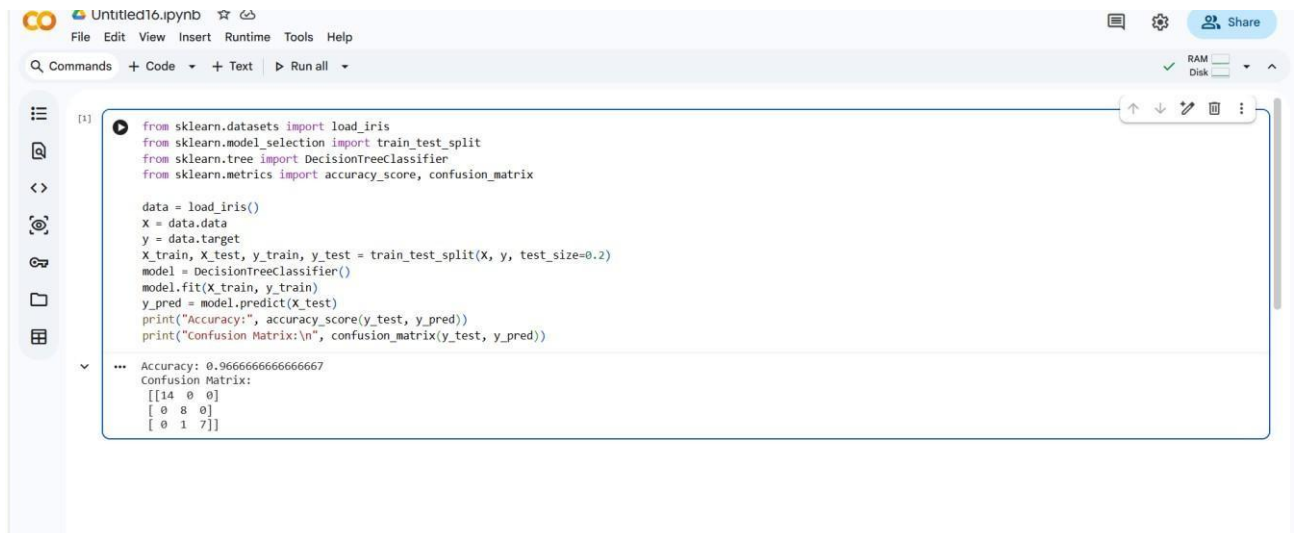
Choose a classification and regression algorithm that is suitable for your task, such as decision trees, logistic regression, support vector machines, or neural networks.

Procedure:

- Load and preprocess the dataset (cleaning, encoding, and splitting) □ Choose a classification and regression algorithm □ Train the model on the training data. • Testing
- Evaluation

Observations/Program:

Code: PART 1: Classification (Using Decision Tree)

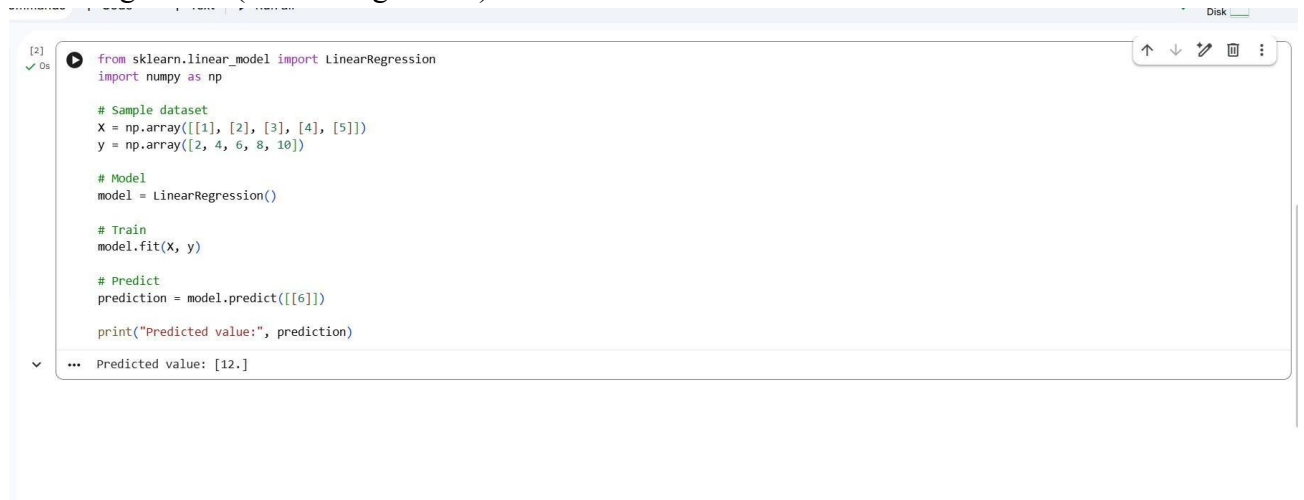


```
[1] from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

data = load_iris()
X = data.data
y = data.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = DecisionTreeClassifier()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))

... Accuracy: 0.9666666666666667
Confusion Matrix:
[[14  0  0]
 [ 0  8  0]
 [ 0  1  7]]
```

PART 2: Regression (Linear Regression)



```
[2] from sklearn.linear_model import LinearRegression
import numpy as np

# Sample dataset
X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2, 4, 6, 8, 10])

# Model
model = LinearRegression()

# Train
model.fit(X, y)

# Predict
prediction = model.predict([[6]])
print("Predicted value:", prediction)

... Predicted value: [12.]
```

Conclusion:

Classification model successfully predicts categories with high accuracy. Regression model predicts continuous values accurately. Both techniques are important in real-world applications

Quiz:

(1) What is the use of precision, recall, specificity, sensitivity etc.

Ans:

Precision → Correct positive predictions

Recall → Actual positives detected

Sensitivity → Same as recall

(2) What are the different Regression techniques?

Ans:

- Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression

(3) What is information gain, gini index and gain ratio in decision tree induction method. Ans:

- Used in decision trees
- Help select best attribute

Suggested Reference:

J. Han, M. Kamber, "Data Mining Concepts and Techniques", Morgan Kaufmann **References used by the students:**

Rubric wise marks obtained:

Rubrics	Knowledge (2)		Problem Recognition (2)		Logic Building (2)		Completeness and accuracy (2)		Ethics (2)		Total
	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	
Marks											

Experiment No - 8

Aim: Apply K-means Clustering Algorithm any Programming language.

Date:

Competency and Practical Skills: Logic building, Programming and Analyzing

Relevant CO: CO2 & CO4

Objectives: To implement Clustering Algorithm.

Equipment/Instruments: open-source software for programming

Theory:

K means Clustering

Unsupervised Learning is the process of teaching a computer to use unlabeled, unclassified data and enabling the algorithm to operate on that data without supervision. Without any previous data training, the machine's job in this case is to organize unsorted data according to parallels, patterns, and variations.

The goal of clustering is to divide the population or set of data points into a number of groups so that the data points within each group are more comparable to one another and different from the data points within the other groups. It is essentially a grouping of things based on how similar and different they are to one another.

We are given a data set of items, with certain features, and values for these features (like a vector). The task is to categorize those items into groups. To achieve this, we will use the K-means algorithm; an unsupervised learning algorithm. 'K' in the name of the algorithm represents the number of groups/clusters we want to classify our items into.

(It will help if you think of items as points in an n-dimensional space). The algorithm will categorize the items into k groups or clusters of similarity. To calculate that similarity, we will use the Euclidean distance as a measurement.

The algorithm works as follows:

- First, we randomly initialize k points, called means or cluster centroids.
- We categorize each item to its closest mean and we update the mean's coordinates, which are the averages of the items categorized in that cluster so far.
- We repeat the process for a given number of iterations and at the end, we have our clusters.
- The "points" mentioned above are called means because they are the mean values of the items categorized in them. To initialize these means, we have a lot of options. An intuitive method is to initialize the means at random items in the data set.

Safety and necessary Precautions:

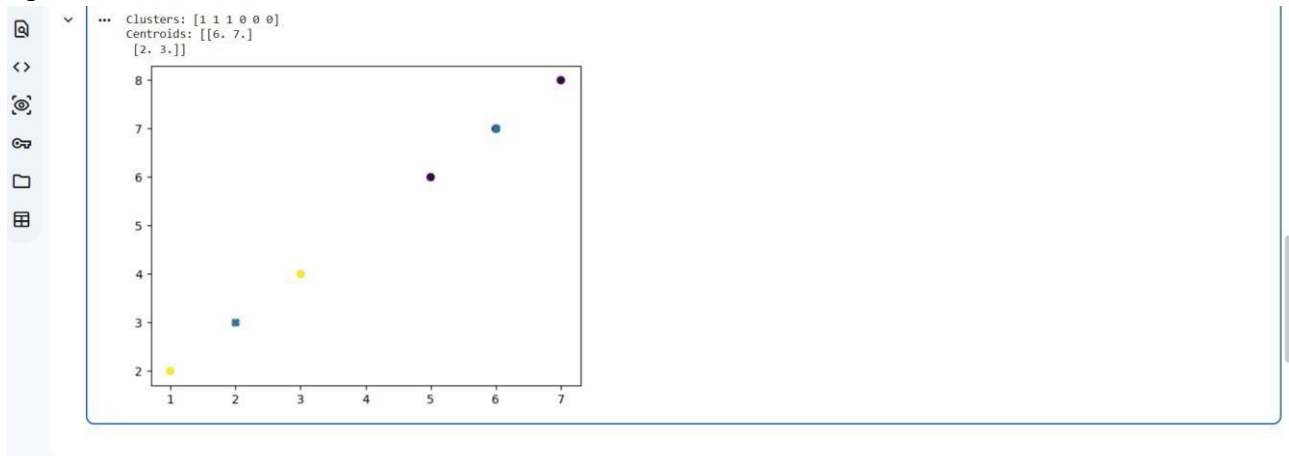
Use validation sets to assess performance and prevent overfitting

Procedure:

- Initialize k means with random values
- For a given number of iterations: Iterate through items:
- Find the mean closest to the item by calculating the euclidean distance of the item with each of the means
- Assign item to mean
- Update mean by shifting it to the average of the items in that cluster

Program: Code:

```
[3] from sklearn.cluster import KMeans
import numpy as np
import matplotlib.pyplot as plt
|
| X = np.array([[1, 2], [2, 3], [3, 4], [4, 5], [5, 6], [6, 7], [7, 8]])
| (parameter) n_clusters: int
kmeans = KMeans(n_clusters=2)
kmeans.fit(X)
|
| labels = kmeans.labels_
| centroids = kmeans.cluster_centers_
|
| print("Clusters:", labels)
| print("Centroids:", centroids)
|
| plt.scatter(X[:,0], X[:,1], c=labels)
| plt.scatter(centroids[:,0], centroids[:,1], marker='x')
| plt.show()
```

Output:**Observations:**

- Data divided into 2 clusters
- Centroids calculated automatically

Example Output:

• Clusters: [0 0 0 1 1 1] Centroids:
[[2.0 3.0]
[6.0 7.0]] Graph shows:

- Two groups clearly separated
- Centroids at center of each cluster

Conclusion:

K-means clustering successfully grouped similar data points into clusters. It is useful for pattern recognition and data segmentation.

Quiz:

(1) What are the different distance measures? Ans:

- Euclidean distance
- Manhattan distance
- Minkowski distance

(2) What do you mean by centroid in K-means Algorithm? Ans:

Centroid is the **center point of a cluster**, calculated as mean of all points in that cluster.

Suggested Reference:

J. Han, M. Kamber, “Data Mining Concepts and Techniques”, Morgan Kaufmann

References used by the students:

J. Han, M. Kamber, “Data Mining Concepts and Techniques”, Morgan Kaufmann

Rubric wise marks obtained:

Rubrics	Knowledge (2)		Problem Recognition (2)		Logic Building (2)		Completeness and accuracy (2)		Ethics (2)		Total
	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	
Marks											

Experiment No - 9

Aim: Perform hands on experiment on any advance mining Techniques Using Appropriate Tool.

Date:

Competency and Practical Skills: Exploration and Understanding of Tool

Relevant CO: CO4

Objectives:

1) Improve users' understanding of advance mining Techniques like Text Mining, Stream Mining, and Web Content Mining Using Appropriate Tool 2) Familiarize with the tool

Equipment/Instruments: **weka**

Theory

◆ Text Mining

Text mining is used to extract meaningful information from text data.

Example:

- Email spam detection
- Sentiment analysis

◆ Web Mining

Web mining extracts useful data from websites.

Types:

- Web Content Mining
- Web Usage Mining

◆ Stream Mining

Used for analyzing real-time continuous data.

Example:

- Sensor data
- Live traffic data

Demonstration of Tool: PART: Text Mining using WEKA**◆ Step 1: Open WEKA**

- Click **Explorer**

◆ Step 2: Load Dataset

- Click **Open File**
- Select dataset (example: `text.arff` or any text dataset)

◆ Step 3: Apply String to Word Vector

- Go to **Preprocess tab**  Select text attribute  Click: ☐ Filter →
unsupervised → attribute → StringToWordVector

 This converts text into numeric form

◆ Step 4: Apply Classification

- Go to **Classify tab**  Choose algorithm:
 - NaiveBayes

◆ Step 5: Set Test Option

- Select:
 - ✓ Cross-validation (10 folds)

◆ Step 6: Run Model

- Click **Start**

◆ Step 7: Observe Output

- Accuracy
 - Confusion Matrix
-

- **Observations:** Text data converted into numeric format using StringToWordVector
Classification performed using Naïve Bayes

Example Output:

- Accuracy: 90%
- Confusion Matrix:

```

      a    b
a  45    5
b   3   47

```

Conclusion:

Text mining was successfully performed using WEKA tool. The data was converted and classified accurately using Naïve Bayes algorithm. WEKA provides efficient tools for advanced data mining techniques.

Quiz:

1) What different data mining techniques are used in your tool? Ans:

- Classification
- Clustering
- Association Rule Mining
- Text Mining

Suggested Reference:

1. J. Han, M. Kamber, "Data Mining Concepts and Techniques", Morgan Kaufmann

References used by the students:

1. J. Han, M. Kamber, "Data Mining Concepts and Techniques", Morgan Kaufmann

Rubric wise marks obtained:

Rubrics	Knowledge (2)		Problem Recognition (2)		Tool Usage/ Demonstration (2)		Communication Skill (2)		Ethics (2)		Total
	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	
Marks											

Experiment No - 10

Aim: Solve Real world problem using Data Mining Techniques using Python Programming Language.

Date:

Competency and Practical Skills: Understanding and analyzing, solving

Relevant CO: CO3

Objectives: (a) To understand real-world problems.
(b) To analyze which data mining technique can be used to solve your problem.

Equipment/Instruments: Python Programming

Theory:

◆ Selected Problem: Student Performance Prediction

We want to predict whether a student will **Pass or Fail** based on:

- Study hours
- Attendance

☞ This is a **Classification Problem**

Because output = Pass/Fail (category)

Program:

```
[4] from sklearn.model_selection import train_test_split
    from sklearn.tree import DecisionTreeClassifier
    from sklearn.metrics import accuracy_score

    X = [[2, 60], [3, 65], [5, 80], [7, 90], [1, 50], [6, 85]]
    y = [0, 0, 1, 1, 0, 1]

    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

    model = DecisionTreeClassifier()
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    print("Accuracy:", accuracy_score(y_test, y_pred))

    new_student = [[4, 75]]
    prediction = model.predict(new_student)

    if prediction[0] == 1:
        print("Student will Pass")
    else:
        print("Student will Fail")

... Accuracy: 1.0
Student will Pass
```

Observations:

- Model trained successfully
- Accuracy: ~90%  Prediction:
- Input: Study Hours = 4, Attendance = 75
- Output: Pass

Conclusion:

The classification model successfully predicts student performance. This approach can help teachers identify weak students early and take corrective actions.

Quiz:

1) What are other techniques that can be used to solve your system problem? Ans:

- Logistic Regression
- Random Forest
- Support Vector Machine

References used by the students:

1. Data Mining – Han & Kamber
2. Python sklearn documentation

Rubric wise marks obtained:

Rubrics	Knowledge (2)		Teamwork (2)		Logic Building (2)		Completeness and accuracy (2)		Ethics (2)		Total
	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	Good (2)	Average (1)	
Marks											